MOHAMAD ASYRAF BIN MOHAMAD ZAILANI

DEGREE-QUALIFIED ENGINEER | ROBOTICS & AEROSPACE SYSTEMS

mohdasyraf999@gmail.com | linkedin.com/in/md-asyraf | asyrafbmz.github.io

NOTABLE PROJECTS | CCA

Anti-Riot Drone - Capstone Project 2025

- An individual project where I designed and built a modular S500 drone to help riot-control officers monitor crowds, give warnings and release small payloads from a safer distance.
- Created a quick-change system that allowed the camera, LRAD speaker and payload dropper to be swapped in under 30 seconds.
- Utilised a Pixhawk flight controller, Ardupilot, Raspberry Pi, Python software, custom circuits and 3D-printed parts.
- Presented the project to SIT faculty, industry guests, and representatives from the Ministry of Home Affairs.

Youth Action Challenge 2020 - EcoRoad 2020

- Led a three-member team to develop EcoRoad, a prototype that used heat from roads to generate electricity and help reduce heat buildup.
- Managed the project planning, prototype development and presentation during the six-month challenge.
- Won the Most Innovative Award, received a grant of up to S$30,000 and placed among the top 12 finalist teams nationwide.

Robotics Club 2018 (ITE College West) 2017 - 2018

- Held the role of Vice President of the club
- Helped plan club activities, prepare members and participate in multiple engineering robotics competitions.

COMPETITIONS PARTICIPATED UNDER ROBOTICS CLUB

Shell Eco Marathon Asia 2018 & 2017

Project: ElectroLiTE – Created an energy efficient electric car built for the Urban Concept category.

- Joined the Electrical Team in 2017
- Held the role of **Team Manage**r in 2018.
- **Represented Singapore** and placed 10th in 2018.

Lee Kwan Yew Technological Award 2017

Project: Roadmaster – Converted road heat into electricity to power a cooling system beneath the road.

- Shortlisted to present the project to the judging panel.
- Received a Certificate of Participation.

Energy Innovation Challenge 2017

Project: Roadmaster – A road-cooling prototype designed to reduce heat buildup in cities.

- Won **1st place** among 23 ITE and polytechnic teams.

GreenWave 2017

Project: Roadmaster – A road-cooling prototype designed to reduce heat buildup in cities.

- Participated using the EIC-winning project for the environmental competition.

Singapore Robotic Games 2017 & 2018

- 2017 – RC Underwater Robot Challenge: Competed with a robot designed to collect and deliver underwater cargo(marbles).
- 2018 – RC Sumo Competition: Competed with a robot designed to push its opponent out of the arena.

Singapore Junior Water Prize 2017

Project: Aqua Legatus – Proof of concept water-saving device to reuse washing-machine water for toilet flushing.

- Presented the project as a solution for reducing household water use.

Sustainable Lifestyle Challenge 2017

Project: Aqua Legatus – Proof of concept water-saving device to reuse washing-machine water for toilet flushing.

- Presented the project and placed 2nd in the competition.

Singapore Amazing Machine Competition 2016

Project: Rube Goldberg machine – A machine where one action triggers the next to complete a simple task.

- Designed and built the machine to with the end result of opening a jar of marbles

EDUCATION

Singapore Institute of Technology - SIAEC 2023 - 2026

- Bachelor of Engineering (Hons w Merit) in **Aircraft Systems Engineering** - GPA: 3.67

ITE College West 2016 - 2018

- Higher Nitec w Merit in **Mechatronics Engineering** - GPA: 3.98 with 6 module distinctions

Ngee Ann Polytechnic 2018 - 2021

- Diploma w Merit in **Aerospace Electronics** - GPA: 3.77 with 2 module distinctions

WORK EXPERIENCE

Aircraft Maintenance Licensed Trainee (B2 Engineer) - Integrated Work Study Programme (IWSP) — **Jan 2026 - Aug 2026**

Singapore Airlines Engineering Company (SIAEC)

- Enrolled in the Integrated Work Study Programme (IWSP) as an Avionics (B2) Licensed Aircraft Engineer (LAE) trainee, undergoing preparatory training prior to operational on-the-job attachment.
- Building understanding of the roles, responsibilities, and professional standards expected of LAEs in an operational airline environment.

Research Assistant (Part-Time / Ad Hoc) — **Feb 2024 – Dec 2025**

Singapore Institute of Technology, Singapore

- Contributed to research and prototype development involving 3D-printed insoles.
- Designed and prototyped a water droplet generator for research applications, involving component design, fabrication, assembly and testing.

Systems & Application Support Developer (Project-Based) — **2021, 2023 - Present**

A2G Music Centre Pte Ltd, Singapore

- Worked in a team to design and develop a modern student database management system, replacing a legacy Windows XP-dependent platform used across multiple branches.
- Built and deployed the application using C#, Windows Forms, and SQL LocalDB.
- Provide ongoing maintenance, feature updates, and bug fixes based on operational needs and user feedback.

Ground Response Force Officer (GRF Officer) — **2021 - 2023**

Singapore Police Force (SPF) | Bukit Panjang NPC

- Served as a Ground Response Force officer at the Singapore Police Force (SPF) during National Service (NS), holding the rank of Sergeant (SC/SGT).
- Responded to '999' emergency calls as a first responder, conducting proactive patrols, lodging reports for members of the public, and supporting major public event deployments sucessfully.

AWARDS & ACHIEVEMENTS

Operational Excellence Award (SPF) — **Dec 2022**

- For excellent efforts and coordination, leading to the swift arrest of one subject for voluntarily causing hurt by dangerous means and abuse against a public servant
- For a well-coordinated operation resulting in arrest of six subjects for rioting with deadly weapons
- For effective management of a large group of subjects involved at a gathering and dealing with public entertainment and liquor license breaches

Merit Award for IKEA Young Designers Award 2019 — **Jul 2019**

- Led a zero-waste product design from concept to presentation, placing in the Top 5 of 132 teams and receiving the People's Choice Award.

Student Commendation Award — **May 2018**

- For winning 10th place in the Shell Eco-Marathon 2018 (Urban Concept Battery Electric category) as the Team Leader

Certificate of Technical Inspection — **Mar 2017 & 2018**

- In recognition of exceptional work in vehicle design, fabrication, testing, and successful completion of the Technical Inspection for Shell Eco-Marathon 2017 & 2018

Tan Kah Kee Young Inventors Award 2018 — **May 2018**

- Presented an engineering idea in a national competition for original inventions and practical solutions.

Champion for Energy Innovation Challenge 2017 — **Jan 2017**

- Won the First Prize for creating a prototype that can reduce Urban Heat Island Effect.

Diamond Service Star Award 2017 — **Jan 2017**

- Recognised for active service and contributions to student activities, CCA events and the community.

SKILLS & COMPETENCIES

Technical Skills

- C#, Arduino, SQL, C, Python, PLC, PX4, Ardupilot

Engineering & Prototyping

- SolidWorks (CSWA Certified)
- Engineering documentation and technical drawings
- Circuit design and soldering
- 3D printing, designing and prototyping
- Manage Work At Height (WAH)

Software & Media Tools

- Microsoft Office
- Adobe Premiere Pro (Basic)
- Adobe Photoshop (Basic)
- Microsoft Visual Studios